

Classical Problems Characterizing the Power of Quantum Computing

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Talk is based on

A single-shot measurement of the energy of product states in a translation invariant spin chain can replace any quantum computation, [quant-ph/0710.1615](#)

A Simple PromiseBQP-complete Problem, *Theory of Computing*, Volume 3, Article 4, 2007

A PromiseBQP-complete String Rewriting Problem, [quant-ph/0705.1180](#)

Challenges in Quantum Computer Science

designing new efficient quantum algorithms

understanding the computational power and limitations of quantum computers in comparison to those of classical computers

Language/Decision Problem

A language L is a subset of $\{0, 1\}^*$.

The corresponding decision problem (“language recognition”) consists in determining if $x \in L$ or $x \notin L$ for given $x \in \{0, 1\}^*$.

Complexity Class BQP

BQP is the class of decision problems that can be solved by a *uniform* family of quantum circuits $\{U_n\}$ in the following sense:

Let $\mathbf{x}$ be a binary string of length n . Then the application of U_n to $|\mathbf{x}, 0\rangle$ produces the state

$$U_n|\mathbf{x}, \mathbf{0}\rangle = \alpha_{\mathbf{x},0}|0\rangle \otimes |\psi_{\mathbf{x},0}\rangle + \alpha_{\mathbf{x},1}|1\rangle \otimes |\psi_{\mathbf{x},1}\rangle$$

such that

$$|\alpha_{\mathbf{x},1}|^2 \geq 2/3 \text{ if } x \in L$$

$$|\alpha_{\mathbf{x},1}|^2 \leq 1/3 \text{ if } x \notin L$$

Measurements of Observables

a measurement of the observable A with spectral decomposition

$$A = \sum_{\lambda} \lambda Q_{\lambda}$$

is a quantum process that

outputs the eigenvalue λ

with probability $\text{tr}(Q_{\lambda} \rho)$

when applied to an arbitrary quantum state ρ

Remarks: Measurements of Observables

note that

the post-measurement state is irrelevant

a quantum process with the property that the expectation of the generated values is equal to $\text{tr}(A \rho)$ may not suffice

Spectral measure

the spectral measure induced by the observable A and the state ρ is the probability distribution on $\Omega = \mathbb{R}$ where

$$\Pr(\lambda) = \text{tr}(Q_\lambda \rho)$$

we set $Q_\lambda = 0$ for all $\lambda \notin \text{Spec}(A)$

a measurement of A in the state ρ allows us to sample according the spectral measure

Accuracy of Measurements

We say that an approximate measurement of A has
maximal error δ and
reliability $1 - \epsilon$

iff

$$\Pr(\lambda \in [a - \delta, b + \delta]) \geq \text{tr}(Q_{[a,b]} \rho) \cdot (1 - \epsilon)$$

for all intervals $[a, b] \subseteq \mathbb{R}$ and all states ρ

where $Q_{[a,b]} = \sum_{\mu \in [a,b]} Q_\mu$

Efficient Implementation of Measurements

let A be a sparse matrix of size N

a measurement of A with maximal error δ and reliability $1 - \epsilon$ can be implemented with resources (time and space) that are polynomial in $1/\delta$, $1/\epsilon$ and $\log N$

the idea is to

simulate time evolution $U \approx \exp(-iA/\|A\|)$

apply phase estimation to U

Quantum Computution with Measurements

there is a family of Hamiltonians $H^{(m)}$ acting on qudit chains $(\mathbb{C}^{60})^{\otimes m}$

$$H^{(m)} = \sum_{j=0}^{m-2} H_{j,j+1}$$

such that a single-shot measurement of the energy of canonical basis states

with error $\delta = 1/\text{poly}(m)$ and reliability $1 - \epsilon$

solves BQP in a probabilistic sense

Properties of the Hamiltonians

the Hamiltonians $H^{(m)}$ are “physically realistic” since

each $H^{(m)}$ is translation invariant

each $H^{(m)}$ consists of nearest-neighbor interactions;

H acts on $(\mathbb{C}^{60})^{\otimes 2}$

Solving BQP in a probabilistic sense

for all input strings x of length n there is

a partition $\mathbb{R} = Y_m \dot{\cup} N_m$ and

a canonical basis state $|\psi_x\rangle \in (\mathbb{C}^{60})^{\otimes m}$

such that

$$\Pr(\lambda \in Y_m \mid x \in L) \geq p_{x,1}(1 - \epsilon)$$

$$\Pr(\lambda \in N_m \mid x \notin L) \geq p_{x,0}(1 - \epsilon)$$

where λ is the random variable defined by the energy measurement of $H^{(m)}$ in the state $|\psi_x\rangle$ and $m = \text{poly}(n)$

the objects m , $|\psi_x\rangle$, Y_m , and N_m can be efficiently computed from x and $U^{(n)}$

Idea of the Proof

let $U = U_r U_{r-1} \cdots U_1$ be an arbitrary quantum circuit

define the non-unitary gates $R := |1\rangle\langle 1|$ and $E := 0$

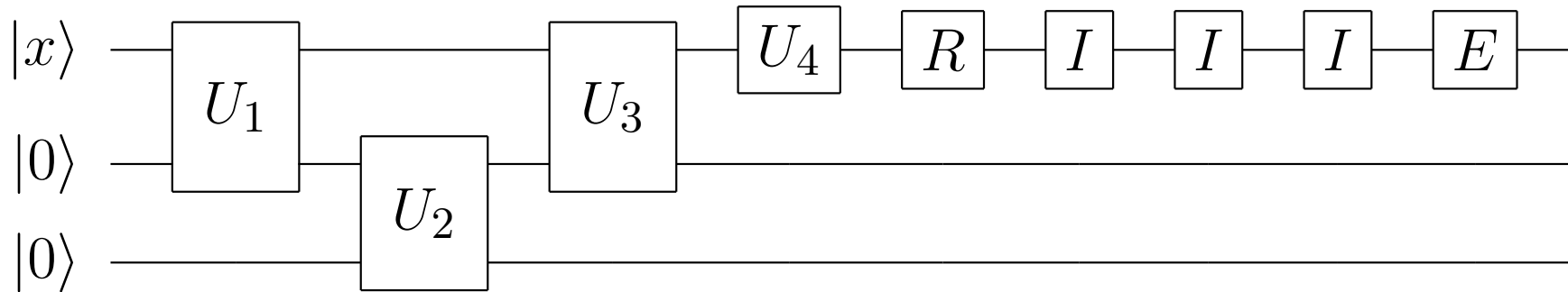
let $s > r$ and define the non-unitary circuit

$$\hat{U} = \hat{U}_{s+1} \hat{U}_s \hat{U}_{s-1} \cdots \hat{U}_{r+1} \hat{U}_r \hat{U}_{r-1} \cdots \hat{U}_1$$

the gates of $\hat{U}$ are defined as follows

$$\begin{aligned}\hat{U}_t &:= U_t && \text{for } t = 1, \dots, r \\ \hat{U}_{r+1} &:= R \\ \hat{U}_t &:= I && \text{for } t = r+2, \dots, s \\ \hat{U}_{s+1} &:= E\end{aligned}$$

Example



$r = 4$ and $s = 8$

read-out gate $R := |1\rangle\langle 1|$ and end-gate $E := 0$

Forward-Time Operator

adjoin a Hilbert space $\mathcal{H}_{\text{clock}}$ to the quantum register

define the forward operator F by

$$\begin{aligned} F\left(|\varphi\rangle \otimes |t\rangle\right) &= \hat{U}_{t+1}|\varphi\rangle \otimes |t+1\rangle \text{ for } t = 0, \dots, s \\ F^\dagger\left(|\varphi\rangle \otimes |0\rangle\right) &= 0 \end{aligned}$$

define the initial state

Intuitive Action of F

F operates as follows:

F implements $U = U_r U_{r-1} \cdots U_1$

F checks the output of U (with read-out gate R)

output NO $\Rightarrow F$ stops

output YES $\Rightarrow F$ carries out idle steps

F stops unconditionally (end-gate E)

Diagonal Entry Estimation

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String Rewriting